

PRINCE CHAUHAN

QUANTITATIVE RESEARCHER | SYSTEMATIC DERIVATIVES & STATISTICAL ARBITRAGE

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SEBI Registered Research Analyst

PROFESSIONAL SUMMARY

Quantitative Researcher and Systematic Derivatives Strategist with 9+ years of experience developing, validating, and deploying options volatility and statistical-arbitrage models across Indian (NSE) and US benchmark derivatives. Specializes in volatility surface spline modeling, Johansen cointegration, market-regime classification, and microstructure-aware execution using point-in-time Python research architectures incorporating explicit transaction costs, STT, fill slippage, margin constraints, walk-forward optimization, and Monte Carlo stress testing.

PROFESSIONAL EXPERIENCE

SMC Global Securities Ltd. — Senior Quantitative Analyst, Systematic Strategies & Quant

Dec 2024 - Present

- Develop causal 1-minute execution loops and point-in-time research pipelines for automated intraday and multi-day options strategies across Nifty, Bank Nifty, Sensex, and single-stock derivatives.
- Architect Bank Nifty basket statistical-arbitrage models using Johansen cointegration, dynamic beta hedging, spread Z-scores, and systematic theta capture across heavyweight constituent equities.
- Build and evaluate systematic models achieving 2.5+ out-of-sample Sharpe and 2.1+ Profit Factor in chronological backtest simulations incorporating explicit 0.50% turnover friction and realistic fill slippage.
- Engineer asynchronous telemetry and data caching using Python Asyncio, broker WebSockets, and memory-mapped (mmap) tick storage, reducing market crossing costs by 1.8 bps and eliminating API rate-limit bottlenecks.
- Structure real-time portfolio Greeks (Delta, Gamma, Vega, Theta), parametric and historical VaR, automated kill switches, and SPAN cross-margin risk models calibrated for institutional notional capacity.

Smart Bulls Capital — Research Analyst, Derivatives & Quantitative Research

Mar 2017 - Nov 2024

- Structured and evaluated systematic multi-leg options volatility strategies (calendar spreads, ratio butterflies, iron condors) with dynamic delta and vega rebalancing driven by real-time IV surface spline fitting.
- Formulated Black-Scholes pricing models, implied volatility surface spline fitters, and real-time dealer gamma exposure (GEX) dashboards to track institutional liquidity concentration.
- Conducted fundamental valuation, financial statement forensics, and DCF modeling across core financial and commercial sectors to support proprietary desk capital allocation.
- Managed desk risk parameters, daily P&L attribution, margin monitoring, and execution tracking across multi-strike options books.

SELECTED QUANTITATIVE RESEARCH & SYSTEMS

- Point-in-Time Institutional Backtesting Framework:** Built modular Python engine utilizing 7+ years of causal, point-in-time 1-minute historical data with zero lookahead bias, incorporating walk-forward optimization, parameter-stability plateaus, explicit 0.50% turnover friction, and 1,000-path Monte Carlo stress testing.
- Bank Nifty Basket Cointegration Model:** Developed statistical-arbitrage model applying Johansen tests, dynamic beta hedging, and rolling spread Z-scores to capture mean-reverting basis differentials and systematic theta across major banking constituents.
- Market Regime Allocation Model:** Designed unsupervised K-Means classifier utilizing rolling IV/RV ratios, price velocity, return skewness, and order-flow imbalance; dynamically scaled short-gamma exposure during volatility breakouts, reducing simulated maximum portfolio drawdown by 28%.

TECHNICAL & QUANTITATIVE SKILLS

- Quantitative Research:** Statistical Arbitrage, Johansen Cointegration, Kalman Filtering, Dynamic Beta Hedging, Volatility Surface Modeling (SVI), Implied & Realized Volatility (IV/RV), Market Regime Detection (K-Means), Walk-Forward Optimization, Monte Carlo Simulation.
- Options & Risk:** Black-Scholes Pricing, Portfolio Greeks (Delta, Gamma, Vega, Theta), Delta/Vega Hedging, Gamma Risk Profiling, Volatility Dispersion, SPAN Cross-Margin Optimization, Historical & Parametric VaR, Dealer Gamma Exposure (GEX).
- Programming & Systems:** Python (Polars, DuckDB, NumPy, Pandas, SciPy, Statsmodels, scikit-learn, VectorBT), SQL, Apache Parquet, REST APIs, WebSockets, Asyncio, Memory-Mapped IPC (mmap), Git, Linux/Bash.
- Markets & Microstructure:** NSE Equity Derivatives (Nifty, Bank Nifty, Sensex), Stock Options & Futures, CBOE SPX/VIX Benchmarks, Market Microstructure, Smart Limit-Order Slicing, Fill Slippage Minimization.

EARLIER EXPERIENCE & EDUCATION

Indian Oil Corporation Ltd. (IOCL) — Corporate Finance Executive

Mar 2016 - Mar 2017

- Corporate treasury cash-flow risk modeling, multi-period corporate valuation, and debt-servicing sensitivity projections.

CA Matta & Associates — Finance Executive

Mar 2014 - Feb 2016

- Financial statement forensics, capital structure valuation modeling, and solvency analysis for institutional clients.
- Bachelor of Commerce (Honours):** Kirori Mal College, University of Delhi (2011 - 2014) | 71% Marks (First Division)
- Chartered Accountancy (CA Final):** Institute of Chartered Accountants of India (ICAI, 2018) | 78% Marks (Cleared Group 1)
- Regulatory Licenses:** SEBI Registered Research Analyst | NISM Series-XV: Research Analyst Certified